

Paper XI — One Currency: Mass, Energy, Light, and Gravity as a Single Rotation Rate

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Series: “A Universe in a Bottle” — Synthesis paper (XI), drawing together Paper I (light as a rotation rate), Paper II (the paired photon), Paper III (Higgs-free mass), and Paper VII (gravity as a lattice dielectric). It introduces no new construction; it shows that the four objects those papers treat separately are one object read four ways, and that this is what fixes the gravitational source.

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Revision 2 (2026-07-01): re-issued as Paper XI (was Paper XII) following the deprecation of the horizon-free “dielectric black hole” paper (F114→F178) and the consequent renumbering of Papers IX–XIII down by one.

*Revision 1 (2026-06-30): **gravity section re-issued per Findings F178–F183.** The canonical gravitational law is now the **induced Einstein equation** $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$, sourced by the **full** stress-energy tensor. The single impedance-matched dielectric $K = e^{2u}$ — and with it the “couples to energy density T^{00} only” reading — is reclassified as the **vacuum/weak-field (PPN-order) representation** of that law (F178). The “one currency” thesis is retained as the **origin story**: the rotation field’s full stress-energy is what gravitates, and $E = mc^2$ is why mass cannot be a source distinct from that. Strong-field consequences — exact Schwarzschild/Kerr black holes (F183), the two-function GR/TOV interior (F181), pressure-weighted Friedmann cosmology (F182), and gravitational waves at $c_{\text{grav}} = c_{\text{lat}} = 1/\sqrt{3}$ (F180) — are folded into §4 and §6. Canonical SI cell per Finding F107: $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P = 1.0664 \times 10^{-34} \text{ m}$.*

Abstract

The BCC quantum-cellular-automaton model contains a single dynamical quantity: the angular rotation rate Ω of the real field pair $(\mathbf{E}, \mathbf{B})$ per lattice tick. We show that the four things usually treated as distinct — rest mass, energy, light, and gravity — are four readings of that one quantity, and that recognising this fixes *what gravity is allowed to be sourced by*. The argument is built from results already verified elsewhere in the series: rest mass is the intercept $\Omega(0) = \arcsin m$ and the speed of light is the slope $d\Omega/d|\mathbf{k}| = 1/\sqrt{3}$ of *one* generator (F26/F167); the photon is the massless, gapless ($\Omega(0) = 0$) limit of that generator, gauge-protected by the $U(1)$ identity channel (F68/F69/F168); and gravity is an induced (Sakharov-type) renormalisation of Ω , whose canonical law is the **induced Einstein equation** $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$ sourced by the rotation field’s **full stress-energy tensor** (F178). Because mass *is* confined rotation energy, there is no separate rest-mass density for gravity to couple to — the rest-mass-sourced alternative (F50/F52/F62) is excluded by the very fact that light gravitates, bends light, and bends by the Einstein factor 2 (F64). The familiar “couples to energy density T^{00} ” dielectric $K(\mathbf{x})$ (F64/F106) is the **vacuum/weak-field representation** of the full-tensor law — exact for lensing, PPN $\beta = \gamma = 1$, redshift and the origin of G , but superseded inside matter and in the strong field, where the second metric function and the pressure ($3p$) source matter: the interior is GR/TOV (F181), cosmology is the pressure-weighted Friedmann pair (F182), black holes are exact Schwarzschild/Kerr (F183), and gravitational waves propagate at $c_{\text{grav}} = c_{\text{lat}} = 1/\sqrt{3}$ (F180). The “one currency” reading survives as the *origin* of all this: $E = mc^2$ is why the rotation field’s energy-momentum is the whole

gravitational source, with nothing left over to reconcile.

1. The thesis

Unification in this model is not the merging of separate forces under a larger symmetry. It is the observation that the model only ever had one thing to begin with. Every dynamical statement in Papers I–X is a statement about the rotation of the real field pair $(\mathbf{E}, \mathbf{B})$ through an angle Ω per tick. What look like four independent concepts are four questions asked of the single dispersion relation

$$\Omega(\mathbf{k}, m) = \arccos(\sqrt{1 - m^2} u(\mathbf{k})), \quad (1.1)$$

where $u(\mathbf{k})$ is the BCC pure-hop amplitude (F167, T5). The map is:

Reading of Ω	Physical name	Where it lives in (1.1)
intercept $\Omega(0, m) = \arcsin m$	rest mass	the m term (zero-wavenumber eigen-rotation)
value of Ω at given $\mathbf{k}$	energy	the whole function $E = (\hbar/\tau) \Omega$
slope $d\Omega/d \mathbf{k} \rightarrow 1/\sqrt{3}$ $\mathbf{x}$ -dependence $\Omega \rightarrow \Omega/K(\mathbf{x})$	speed of light gravity	the $m = 0, \mathbf{k} \rightarrow 0$ tangent a position-dependent renormalisation of the rate (weak-field reading of the full-tensor law)

This paper makes that table precise and draws the one non-obvious consequence: it fixes what gravity is allowed to be sourced by — the rotation field’s full stress-energy tensor, of which the position-dependent rate $\Omega/K(\mathbf{x})$ is the vacuum/weak-field representation.

2. Mass and energy are the same confined rotation (Papers I, III; F26, F167)

Paper III builds rest mass from the chiral- $SU(2)$ complex-mass step, and F167 closes the structural question the audit raised (G1): the mass term is *forced and unique* — among the sixteen Hermitian Dirac covariants exactly two are simultaneously chirality-off-diagonal and spin-rotation scalars, of which one is the physical mass and the other a pure-gauge phase. Unitarity then forces the kinetic rescale $n = \sqrt{1 - m^2}$ ($n^2 + m^2 = 1$, residual 2×10^{-16}), and the rest map’s eigen-phase is

$$\Omega_{\text{rest}}(m) = \arcsin m, \quad \cos \Omega_{\text{rest}} = \sqrt{1 - m^2} = n. \quad (2.1)$$

The same generator gives $d\Omega/d|\mathbf{k}| \rightarrow 1/\sqrt{3} = c_{\text{lat}}$ at $m = 0$ (F26). Rest mass is the *intercept* and the speed of light is the *slope* of one function — verified together to $< 10^{-12}$ (F167, T5). Restoring the lattice clock, $E = (\hbar/\tau) \Omega$, so

$$E_0 = \frac{\hbar}{\tau} \arcsin m \xrightarrow{m \ll 1} \frac{\hbar}{\tau} m, \quad m_{\text{phys}} = E_0/c^2. \quad (2.2)$$

The conclusion is the linchpin of the whole synthesis: **there is no “mass substance” distinct from energy.** A massive particle at rest is rotation energy that has been confined ($\Omega(0) \neq 0$); a particle in motion carries more of the same rotation. $E = mc^2$ is not a bridge between two categories — it is the single statement that energy and confined rotation are the same thing, read once as a rate and once as an intercept.

3. The photon is the gapless limit of the same generator (Papers I, II, IV; F68, F69, F168)

A photon is what (1.1) becomes when there is no confinement: $m = 0$, so the intercept vanishes and the *only* energy available is the propagating slope term,

$$\Omega_{\text{pair}}(\mathbf{k}) = \omega^+(\mathbf{k}/2) + \omega^-(\mathbf{k}/2), \quad \Omega_{\text{pair}}(0) = 0, \quad \frac{\Omega_{\text{pair}}}{|\mathbf{k}|} \rightarrow \frac{1}{\sqrt{3}}. \quad (3.1)$$

Three structural facts (F168) make this the *forced* form, not an assumption:

1. **Masslessness is the absence of a self-loop.** The BCC walk is a pure body-diagonal hop with no on-site term ($A_0 = 0$), so $u(0) = 1$ and $\omega^\pm(0) = \arccos 1 = 0$. An on-site deficit ε would open a gap $\omega(0) \approx \sqrt{2\varepsilon}$; the photon has none (F168, B1).
2. **The binder is the conserved vector current.** $\omega^-(\mathbf{k}) = \omega^+(-\mathbf{k})$ exactly (residual 0), so the symmetric pair is the parity-even $U(1)$ identity channel that minimal coupling forces (F68); the single-branch alternative is the chiral, birefringent object excluded by GRB/AGN polarimetry (F67).
3. **Zero binding energy is gauge-protected, not tuned.** Mass lives entirely in the branch-coupling vertex X (gap = m , F168 B4); the photon’s identity-channel coupling has $X \equiv 0$, so it cannot generate a mass term and $\Omega(0) = 0$ is forced. This is Weinberg’s massless-spin-1 theorem realised on the lattice, and the sharp contrast with the scalar sibling (F73/F74), whose gap is *unprotected* and needs fine-tuning.

So the photon carries energy with zero mass and zero charge for one reason: it is the parity-even identity channel of a pure-hop walk. Pure hop — no mass intercept; identity channel — no charge, no mass term. The only thing it can carry is the moving rotation, $E = c|\mathbf{k}| \hbar/\tau$ — the model’s reading of $E = pc$. For a photon, “to exist” and “to move” are the same statement.

4. Gravity renormalises the one rate, and the full stress-energy is its source (Paper VII; F178, F64, F106)

If mass, energy and light are all the rotation Ω , then a theory of gravity in this model has exactly one thing to act on: the value of Ω as a function of position and time. Whatever object carries the rotation also carries its energy, momentum, and stress — and because mass *is* confined rotation energy (Section 2), there is no separate rest-mass substance hiding behind it. So the gravitational

source is the rotation field’s own **stress-energy tensor**, in full. The canonical law of the model is the **induced Einstein equation** (F178)

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad G = \frac{a^2 c^3}{8\pi\sqrt{3}\hbar} \quad (\text{structural, F79/F107}), \quad (4.1)$$

with no free coupling: G collapses to pure lattice quantities. This is gravity as an emergent, induced (Sakharov-type) renormalisation of the rotation substrate, whose low-energy dynamics are exactly Einstein’s. The unification still *fixes the source* — it just fixes it to the whole tensor, not to one component.

The rest-mass route is what gets excluded. The fact that prompted the older “couples to energy” reading — that massless photons feel and bend in gravity — is exactly what kills a rest-mass-density source. The model once offered two gravities:

Route	Source	Light bending	Light bends light?
Rest-leg (F50/F52/F62, superseded)	rest-mass density ρ	factor-1 (Newton); 0 for pure light	no
Energy/full-tensor (F64→F178, canonical)	stress-energy $T_{\mu\nu}$	factor-2 (Einstein)	yes

Equal-energy rest-mass and massless-field sources bend probe rays equally (ratio 1.00002, F64 D-EM3), and a Maxwell pulse bends through a well sourced by a *massless* energy lump at full factor-2 (F64 D-EM6). Observation says light gravitates and bends by factor 2 — so the rotation field’s energy-momentum, not any rest-mass density, is the source. $E = mc^2$ is why this is consistent: “the full stress-energy gravitates” already contains mass, with nothing left to reconcile.

4.1 The dielectric is the vacuum/weak-field representation (F64, F106)

In the regime where $T_{\mu\nu} \rightarrow 0$ (vacuum) or $p \ll \rho c^2$ (Newtonian), the full-tensor law (4.1) reduces to a single impedance-matched index $K(\mathbf{x})$ renormalising the **(E, B)** rotation rule — the dielectric of Paper VII, with metric legs $A = 1/K$, $B = K$ and the reciprocal lock $AB \equiv 1$. Its static sourcing law (F106),

$$\nabla^2 \ln K(\mathbf{x}) = -\frac{8\pi G}{c^4} T^{00}[\psi] = -\frac{a^2 c_{\text{lat}}}{\hbar c} T^{00}[\psi], \quad (4.2)$$

is exactly the **static weak-field reduction** of (4.1) — F106 derived it as precisely that. A rest eigenstate hands the source $\sqrt{A} m |\Psi|^2 \rightarrow m |\Psi|^2$ — energy, not bare probability (F106, E5). What makes one scalar reproduce GR’s PPN phenomenology — factor-1 redshift *and* factor-2 bending together — is the impedance-matched placement $\varepsilon = \mu = K$: the lock $AB \equiv 1$ keeps $\sqrt{\mu/\varepsilon} = 1$ exactly u -independent, so the rotation stays a *proper* rotation with no scalar contamination (F64 D-EM5). The result is GR-identical at the post-Newtonian level: $\beta = \gamma = 1$, Mercury 42.98”/century, Shapiro delay (F64 D-EM9).

4.2 Where one scalar is not enough (F173, F178, F181)

A single impedance-locked scalar cannot be the *exact* field equation, because $A = 1/K$, $B = K$ is one function where Einstein's equations need two. The exact Einstein tensor of the dielectric metric carries an irreducible *anisotropic* field stress $p_r = -p_t = -e^{-2u}u'^2/8\pi$ and no isotropic matter-pressure term (F173, sympy-exact) — so it omits GR's Tolman $3p$ source. This is null in the solar system ($\sim 10^{-11}$) but $O(3p/\rho c^2)$ — a few to tens of percent — inside neutron stars, and a clean factor of 2 in the radiation era. The full-tensor law (4.1) has no such gap: inside matter the metric carries its **second independent function** (A, B free, $AB \neq 1$) and the dynamics are GR/TOV. The covariant two-function interior kernel (F181) solves the isotropic perfect-fluid Einstein equations exactly where the single scalar forces $p_r = -p_t$, reproducing GR-TOV to the integrator floor, and the F62/F64 dynamic Dirac battery — equivalence principle, redshift $\propto \sqrt{-g_{tt}}$, factor-2 deflection, unitary backreaction — passes unchanged on the two-function background. Nothing in the weak-field phenomenology is lost; only the *exact* strong-field content changes.

4.3 The same light cone for gravity and light (F180)

The causal completion of (4.2) is the hyperbolic wave equation $(\nabla^2 - c_{\text{lat}}^{-2}\partial_t^2) \ln K = -(8\pi G/c^4)T^{00}$ (F180, D-GW). Its speed is not chosen: the dielectric has zero tree stiffness (its conformal factor has no bare kinetic term, F79), so the graviton's inverse propagator *is* the induced vacuum polarisation of the lattice's own rotation modes, and that loop can carry only the constituent light cone $c_{\text{lat}} = 1/\sqrt{3}$. Gravitational waves and photons therefore solve the *same* operator $\square_{\text{lat}} - c_{\text{grav}} = c_{\text{lat}} = c_{\text{photon}}$ is one symbol, not two numbers tuned to agree. The GW170817 residual is bounded at $\sim 3 \times 10^{-83}$, 68 orders below the observed limit. This is the rotation-currency thesis at its sharpest: light and gravity share a light cone because they share the rotation rule.

5. The closed chain

The unification is a single implication chain, each link verified in the cited finding:

$$\underbrace{\text{mass} = \text{confined (E, B) rotation}}_{\text{F26/F167}} \Rightarrow \underbrace{\text{the gravitational source is the rotation field's full } T_{\mu\nu}}_{\text{F178}} \Rightarrow \underbrace{\text{it gravitates light per unit energy as mass, bends it by factor 2}}_{\text{F64 D-EM3/D-EM6}} \Rightarrow \underbrace{\text{the law is the induced Einstein equation } G_{\mu\nu}}_{\text{F178/F181}}$$

In the vacuum/weak field this collapses to the impedance-matched dielectric — one scalar that keeps the rotation proper while reproducing GR's PPN tests (F64 D-EM5). $E = mc^2$ is the hinge that makes the rotation field's energy-momentum the *whole* source, with no rest-mass density to add. Mass, energy, light and gravity are not unified by adding structure; they were one rotation all along, and gravity is simply that rotation's full stress-energy made to curve its own substrate.

6. What this is, and what remains open

This synthesis is a re-reading, not a new result: every equation and number above is quoted from a verified finding (Section 7). Its claim to content is (i) the recognition that the four concepts share

one generator, and (ii) the demonstration that gravity’s source is *forced* — to the rotation field’s full stress-energy — by mass–energy equivalence together with the empirical gravitation of light.

The model is GR-identical at the PPN level ($\beta = \gamma = 1$, Mercury 42.98″/century, factor-2 bending; Paper VII, F64 D-EM9), and under the full-tensor source (F178) its low-energy dynamics are exactly Einstein’s: the canonical black hole is exact Schwarzschild/Kerr — horizon present, GR shadow $3\sqrt{3}M$, Hawking radiation, photon-sphere ringdown — with one genuinely new feature, a lattice-regulated core (curvature saturating at the cell scale, $r_{\text{core}} \sim (r_g^2 a^4)^{1/6} \propto M^{1/3}$) replacing the point singularity (F183). This **supersedes** the horizon-free “dielectric black hole” (deprecated, F114), now understood as a PPN-order representation artifact; the ngEHT +4.6% shadow target is withdrawn. Cosmology is the standard pressure-weighted Friedmann pair, $\ddot{a}/a = -(4\pi G/3)(\rho + 3p/c^2)$ (F182); the demoted energy-only law would have mis-weighted the radiation era by exactly a factor 2.

The model’s distinctive gravitational content therefore narrows, deliberately, from “novel strong-field phenomenology” to **the origin and value of G** — gravity as an induced renormalisation of the rotation substrate — plus the shared light cone $c_{\text{grav}} = c_{\text{lat}}$ (F180). The honest open edges: the Kerr/frame-dragging interior (T_{0i}) on the two-function kernel and a tabulated-EoS TOV for an absolute neutron-star placement (F181); the interacting two-body photon wavefunction — F168 proves *why* the bound pair sits at zero binding energy, but the first-principles Bethe–Salpeter build is unfinished; and the overall mass *scale* (F119 no-go), where the spectrum’s shape is locked but its single normalisation needs an external anchor or a dynamical condensate.

7. Findings and modules cited

Result used	Finding	Module
$c_{\text{lat}} = d\Omega/d \mathbf{k} = 1/\sqrt{3}$ (rotation-rate slope)	F26	ca-simulation/ca_bcc.py
Rest mass = $\arcsin m$ = intercept of the same generator; mass forced & unique	F167	ca_bcc.py, ca_dirac.py
Paired-spinor photon: massless, luminal, non-birefringent	F69	ca-simulation/ca_photon_pair.py
$U(1)$ identity channel forces the even (non-birefringent) photon	F68	ca-simulation/ca_wmu.py
Photon binding gauge-protected ($\Omega(0) = 0$ forced, $A_0 = 0$ pure hop)	F168	(analytical; ca_bcc walk)

Result used	Finding	Module
Canonical law: induced Einstein equation $G_{\mu\nu} = (8\pi G/c^4)T_{\mu\nu}$; dielectric demoted to vacuum/weak-field representation	F178	ca-simulation/ca_gravity.py, ca_stellar.py
Dielectric gravity $K = e^{2GM/rc^2}$, $AB \equiv 1$ (weak-field rep); radiation gravitates as mass; light bends light	F64	ca-simulation/forks/gr_fork_F64_em_conn
$\psi \rightarrow K$ sourcing law $\nabla^2 \ln K = -(a^2 c_{\text{lat}}/\hbar c)T^{00}$ (no free G); static weak-field reduction of F178	F106	tests/findings/test_F106_psi_K_sourcing
Single scalar omits the Tolman $3p$ source ($p_r = -p_t$); null in solar system, $O(3p/\rho c^2)$ in neutron stars	F173	ca-simulation/ca_tolman.py
Covariant two-function GR/TOV interior (A, B free, $AB \neq 1$); F62/F64 dynamic battery unchanged	F181	ca-simulation/ca_interior_metric.py
Pressure-weighted Friedmann cosmology $\ddot{a}/a = -(4\pi G/3)(\rho + 3p/c^2)$; energy-only law off by factor 2 in radiation era	F182	ca-simulation/ca_cosmology.py
Gravitational-wave equation; $c_{\text{grav}} = c_{\text{lat}} = 1/\sqrt{3}$ (GW170817 margin $\sim 10^{-83}$)	F180	ca-simulation/forks/gr_fork_F180_gw_spe
Strong field: exact Schwarzschild/Kerr (horizon, shadow $3\sqrt{3}M$, Hawking, ringdown) + lattice-regulated core	F183	ca-simulation/ca_blackhole.py
Canonical SI cell $a = \sqrt{8\pi} 3^{1/4} \ell_P = 6.5978 \ell_P$; structural G	F107	—
Horizon-free dielectric black hole (superseded by F183)	F114	—

External lineage: Puthoff polarizable-vacuum and Ostoma–Trushyk dielectric-vacuum programmes (the energy-coupled-index line, now understood as the vacuum/weak-field representation of induced Einstein gravity); see [references/ostoma-trushyk-1999-summary.md](#).

End of Paper XI.
